Mallika Ramesh Chourasia

chourasiamallika5@gmail.com | (312)-871-8010 | **Portfolio** | **LinkedIn**

Education

University of Illinois Chicago *Aug 2024 – May 2026*
Master of Science in Management Information Systems GPA: 3.92/4

University of Mumbai *Aug 2020 – May 2024*
Bachelor of Technology in Electronics and Telecommunication Engineering GPA: 3.78/4

University of Mumbai *Aug 2022 – May 2024*
Honors degree in Artificial Intelligence and Machine Learning GPA: 3.86/4

Professional Experience

UIC Office of Technology Management, Chicago, IL | Data Analyst | Sept 2025 – Present

- Built 10+ Power BI dashboards on SQL Server using DAX and Power Query for 25+ users across 3 departments, standardizing KPI definitions so monthly correction requests dropped by 20%
- Automated repetitive data pulls across 6+ source systems with Python scripts and SQL Server stored procedures, saving 10+ hours monthly and shortening request turnaround by 60%, from 5 days to 2
- Wrote SQL data-quality checks with window functions to reconcile records across systems, tracing 15+ recurring discrepancies to their source and cutting manual cleanup before each report by 30%

UIC MarTech Hub, Chicago, IL | Research Assistant | Sept 2025 – May 2026

- Segmented 50K+ Salesforce and HubSpot contacts using cohort and regression analysis in R, identifying where prospects dropped out of the funnel and refocusing targeting, which raised email clicks by 28%
- Developed ARIMA time-series models on Databricks to forecast monthly event and webinar sign-ups, tracking each experiment in MLflow and reducing forecast error by 35% on held-out months
- Evaluated 4 automation platforms (HubSpot, Salesforce, n8n, Make) on cost and fit, then set up daily Apache Airflow DAGs that replaced week-old HubSpot exports with same-day campaign data

Milestone Industries, India | Junior Data Analyst | Feb 2023 – May 2024

- Analyzed 500K+ daily sensor records across 4+ kitchen appliance production lines in SQL and Python, flagging temperature and vibration anomalies in Power BI drill-downs that cut investigation time by 55%
- Trained Scikit-learn classifiers on equipment telemetry to predict machine failures, tuning alert thresholds to balance missed failures against false alarms, which extended mean time between failures by 40%, from 15 to 21 days
- Stabilized nightly sensor-data loads by adding health checks, automatic retries, and failure alerts, so recurring job failures were resolved before morning production reports reached plant supervisors

Winner Impex, India | Business Analyst Intern | July 2022 – Dec 2022

- Optimized reorder points for 2,000 active SKUs across books, stationery, and school supplies by building SQL and Excel demand-variance models with the procurement team, increasing annual inventory turnover by 25%
- Diagnosed the order-processing bottleneck behind recurring supplier delays and launched a supplier-tracking workflow in Power Automate and Power BI, reducing the pending-order backlog by 30%

Projects

Uber NYC Pickups Data Pipeline & Metrics API | *Python, PostgreSQL, FastAPI, Pytest, GitHub Actions* | **Link**

- Engineered an ingestion pipeline that loads 4,534,327 Uber NYC pickups (Apr–Sep 2014) into PostgreSQL in batches, applying 5 validation rules, flagging 82,581 duplicate rows (1.8%), and revealing a Thursday 5 PM demand peak
- Deployed 6 FastAPI REST endpoints for pickup metrics, covered by 51 Pytest tests running in GitHub Actions CI, and added database indexes that cut lookup latency by 97%, from 61 ms to 1.7 ms

TravelBuddy: Generative AI Trip Planner | *AWS Bedrock, Flask, Docker, EC2, Amazon QuickSight*

- Launched a generative AI trip planner that turns each traveler's budget and dates into day-by-day itineraries, running AWS Bedrock behind a Flask app in Docker on EC2, with 16,000+ itineraries generated
- Tracked first-time and returning users in Amazon QuickSight dashboards, feeding user feedback and logs into prompt revisions that improved itinerary quality with each release

Opening the Conversation: Analysis of Men's Mental Health | *PyTorch, Hugging Face, SHAP, LIME* | **CIST 2026 London**

- Fine-tuned RoBERTa with PyTorch and Hugging Face Transformers to classify 120K+ Reddit and YouTube posts on men's mental health, then used hypothesis tests to compare trends across demographic groups over time
- Explained model predictions with SHAP and LIME, identifying the language patterns that most influenced each classification, and documented the full methodology in a paper for CIST 2026

Skills

Programming & Querying: Python, SQL, R, Java, Bash, CTEs & Window Functions, Stored Procedures, Query Optimization
Data Engineering: ETL/ELT Pipelines, Apache Spark, Kafka, Apache Airflow, dbt, Data Modeling, Dimensional Modeling
Cloud & Warehouses: AWS, Snowflake, BigQuery, Amazon Redshift, Databricks, PostgreSQL, SQL Server, Azure, GCP
BI & Visualization: Power BI, Tableau, Looker, DAX, Power Query, SSRS, Dashboard Design, Semantic Layers
Analytics & Statistics: Exploratory Data Analysis, A/B Testing, Regression, Time-Series Forecasting, Hypothesis Testing
Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, Hugging Face, NLP, Generative AI, LLM APIs
Business Analysis & Tools: Requirements Gathering, Functional Specs, Process Mapping, Data Quality, Git, Jira, Agile

Leadership

INFORMS UIC Chapter | Director of Marketing | May 2025 – May 2026

- Led social media strategy and content for the INFORMS UIC chapter across 3+ platforms, using targeted outreach on each to grow workshop attendance by 29%, from 70 to 90 students